

COVER-KBC

Coverage-guided Open-set Verification and Elicitation for Relation-wise Knowledge Base Construction

Technical Method and Architecture Proposal for AKBC Shared Task 2026

Central thesis – upgraded version

Treat each subject–relation pair as a specialist-routed *active set discovery* process. The system does not merely generate objects: it compiles each relation into a prompt program, performs “parametric retrieval” through structurally independent branches, maintains atomic evidence, measures uncertainty from logits and disagreement, estimates what remains undiscovered, and allocates test-time compute to the appropriate module until the expected marginal benefit becomes too small.

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Technical concept note – architecture evolution plan, not the final 6-page system paper

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1 Executive summary

AKBC Shared Task 2026 requires predicting the complete set of object strings for a subject s and relation r ; the number of objects may be zero, one, or very large. The official setting is closed-book: no web search, no RAG, no external factual corpus, no KB lookup, and no additional training; agentic/multi-step inference and non-neural processing are allowed, while the total number of neural parameters used at inference must not exceed 32B [1]. The previous proposal established the right foundation: Relation Contract, multi-view discovery, calibrated label logits, an Evidence Graph, blind verification, residual coverage, active scheduling, and adaptive stopping. That thesis is preserved. This proposal extends it into a layered architecture with specialist modules. The original document framed the task as active set discovery with evidence and control; the upgraded version strengthens routing, prompt compilation, specialist branches, and test-time control without changing the scientific core.

Current empirical status

The real system has already run end-to-end with frozen open-weight models and no fine-tuning. The current profile uses Mistral-Small-3.2-24B-Instruct-2506 as enumerator and Qwen3.5-4B as verifier/cross-family checker, with an audited published-parameter total of about 28.67B. Full validation currently reaches macro-F1 **0.4230**; per-relation results are highly uneven: borders are close to saturation, whereas capacity and awards remain major bottlenecks. This motivates the transition from a single shared pipeline to **relation-specialist routed inference**.

Major upgrades in COVER-KBC Super-System:

1. **Layered specialist routing**: retain M0–M8 and add M9–M21 to route each relation/query to numeric, large-open-set, null/temporal, or small-set specialists.
2. **Prompt Program Compiler**: prompts are no longer isolated hand-written strings; every relation has a keyword specification, positive/negative semantics, a branch plan, output grammar, and verification plan. Offline prompt search follows ideas from APE, OPRO, MIPRO, AMPO, and black-box prompt optimization without updating model weights [13, 14, 15, 16, 17].
3. **RAG-inspired but challenge-legal**: Self-RAG, HyDE, Query2Doc, Rewrite-Retrieve-Read, FLARE, and LLaTrieval are used as sources of ideas for query rewriting, active retrieval, and retrieval verification; there is *no* external retrieval in the submission. COVER’s “retriever” only probes the parametric memory of frozen models through independent prompts/facets, and pseudo-context is never treated as ground-truth evidence [21, 22, 23, 24, 25, 26].
4. **Atomic consensus + specialist verification**: ASC, Self-Consistency, CoVe, and verifier literature are reinterpreted as an atomic evidence graph with independence groups, numeric clusters, set-membership verification, and bidirectional checks [5, 4, 6, 28].
5. **Compute-optimal active control**: Adaptive-Consistency, LATS, SELF-DISCOVER, and inference-scaling work motivate budget reservation, action utility, and micro-lookahead; no learned policy is used [7, 8, 11, 32, 33].
6. **Model-role routing**: within the same parameter budget, large and small models may swap roles by relation; time-sensitive relations can trigger fresh-family cross-recall, while static relations retain short inference paths.

Development principle from this point onward

Every architectural change must correspond to a clearly motivated module/layer and hypothesis. After implementation and audit: run TRAIN to inspect failures, run full VALIDATION, submit to the leaderboard, record $\Delta F1$ and per-relation behavior, and only then keep/remove the module and proceed to the next step. Validation is intentionally used as recurring external feedback in the engineering loop; inner-loop prompt/threshold search should still primarily use TRAIN to reduce leaderboard overfitting.

2 Problem setting, rules, and design consequences

2.1 Official task

For (s, r) , the system predicts

$$\hat{O}(s, r) = \{o_1, o_2, \dots, o_k\}, \quad k \in \{0, 1, 2, \dots\}.$$

String outputs are normalized/alias-matched; the two numeric relations are considered correct within a 5% relative tolerance [1]. The six relations have fundamentally different error surfaces, so a single homogeneous prompt/policy is structurally inappropriate.

Table 1: Task structure and current failure modes.

Relation	Program type	Failure mode to address	Val F1
countryLandBordersCountry	SMALL_SET	completeness + maritime/territory near-misses; already strong	0.9531
personHasCityOfDeath	NULL_SINGLE	living/unknown gate, locality recall, temporal freshness	0.4000
hasCapacity	NUMERIC	severe over-abstention; UNKNOWN currently suppresses recall	0.0900
awardWonBy	LARGE_OPEN_SET	over-generation, missingness, nominee/-work/award confusion	0.1806
companyTradesAtStockExchange	SMALL_SET/TEMPORAL	parent/subsidiary, delisting, primary vs secondary listing	0.5657
hasArea	NUMERIC	unit/definition variants, value consensus	0.3000

2.2 Hard compliance boundary: RAG is an inspiration, not a runtime component

Do not misinterpret the “RAG layer”

The official rules prohibit web search, RAG, external factual corpora, and KB lookup [1]. Therefore this proposal **does not add real RAG**. There is no BM25/vector database over Wikipedia, no search API, no Wikidata lookup, and no retrieval from train labels at test time. Any design that depends on external factual documents is **out of scope** for the official submission.

We reuse only three ideas from the RAG literature:

- **query rewriting/specification:** transform relation + subject into clearer semantic queries/facets;
- **active retrieval scheduling:** only “retrieve” more when uncertainty or coverage gap remains high;
- **retrieval verification:** a candidate/pseudo-context produced by one branch must be checked by an independent branch.

In COVER, the retrieval target is not an external corpus but the **parametric memory of the frozen models themselves**. Query rewriting ideas from Rewrite-Retrieve-Read, Query2Doc, HyDE, and Self-RAG are converted into prompt rewriting/pseudo-memory elicitation; a pseudo-document is only a candidate source, never evidence of truth [24, 23, 22, 21]. FLARE motivates the “ask-more-when-low-confidence” trigger; LLattribution motivates verify-before-trust behavior [25, 26].

2.3 No training, but optimization is still possible

No fine-tuning, LoRA, continued pretraining, learned verifier head, or learned router is used. However, the system may:

- search discrete prompts on TRAIN;
- use deterministic calibration, thresholds, unit conversion, clustering, and scheduling;
- use multiple frozen models as long as the published-parameter total is legal;
- collect logits, entropy, margins, and token costs;
- use independent same-model and cross-model inference calls.

3 Research thesis and objective

3.1 From active set discovery to routed active evidence acquisition

The previous proposal optimized a policy π for F1 and cost. The upgraded version additionally introduces a router ρ , prompt program P_r , and specialist family E_r :

$$(\rho^*, \pi^*, P^*) = \arg \max_{\rho, \pi, P} \mathbb{E} \left[F_1(\hat{O}, O^*) - \lambda C - \mu R_{fp} - \nu R_{miss} \right],$$

where C is compute cost, R_{fp} is false-positive risk, and R_{miss} is residual-missingness risk. No component of ρ, π, P is required to be a learned neural component.

3.2 Extended state

At round t :

$$S_t = (G_t, U_t, R_t, B_t, Y_t, C_t, Q_t, D_t, M_t),$$

where G_t is the Evidence Graph, U_t unresolved candidates, R_t residual coverage, B_t remaining budget, Y_t action yields, C_t calibrated verifier signals, Q_t the query-risk profile, D_t prompt/view disagreement, and M_t model-role state. The action set contains specialist probes, pseudo-memory probes, candidate-free recall, blind verification, counterfactual verification, reverse checks, cross-model checks, optional contrastive decoding, resampling, and STOP.

4 Overall architecture: 8 layers, M0–M21

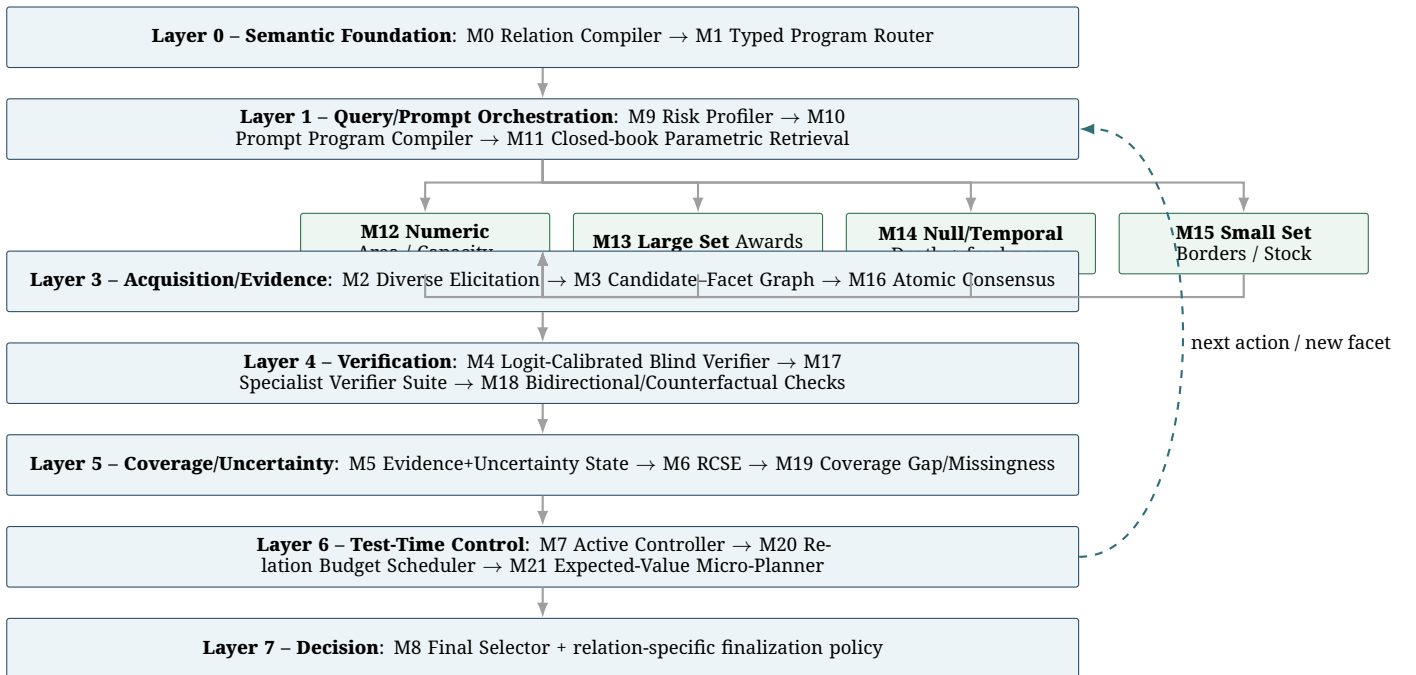


Figure 1: COVER-KBC Super-System: the M0–M8 core is preserved; M9–M21 add prompt orchestration, specialist branches, consensus, and compute control.

Many modules, but a tight paper narrative

Not all 22 modules are novelty claims. The final paper should claim only the core abstractions: **relation-typed active evidence acquisition**, **independence-aware calibrated verification**, and **coverage/uncertainty-guided test-time control**. Specialist modules are executable instantiations demonstrating that the abstraction adapts to genuinely different relation error surfaces.

4.1 Module responsibility map

Table 2: Module map of the upgraded system.

ID	Module	Responsibility	Neural?
M0	Relation Compiler	Exact semantics, types, hard negatives, normalization, allowed actions.	No
M1	Typed Program Router	SMALL_SET / NULL_SINGLE / NUMERIC / LARGE_OPEN_SET.	No
M2	Diverse Elicitation	Structured independent views and candidate acquisition.	Yes
M3	Candidate-Facet Graph	Provenance and signed evidence.	No
M4	Blind Logit Verifier	A/B/C logits, contextual calibration, prompt disagreement.	Yes
M5	Evidence+Uncertainty	F/L/X/C/U state and candidate status.	No
M6	RCSE	Residual search need, not cardinality oracle.	No
M7	Active Controller	Action loop and adaptive stopping.	No
M8	Final Selector	Contract-valid final object set.	No
M9	Risk/Difficulty Profiler	Dynamic risk vector: ambiguity, temporal sensitivity, expected cardinality, candidate explosion, numeric disagreement.	No
M10	Prompt Program Compiler	Compile relation lexicon + branch templates + output grammar + negative constraints.	No
M11	Closed-book Parametric Retrieval	RAG-inspired query rewrite, pseudo-memory and self-ask probes; no external documents.	Yes
M12	Numeric Specialist	Multi-probe numeric acquisition, units, clustering, UNKNOWN-aware decision.	Mixed
M13	Large-Set Specialist	Facet decomposition, atomic set union, shortlist, reserved verification.	Mixed
M14	Null/Temporal Specialist	Existence/death status, locality, freshness branch, null evidence.	Mixed
M15	Small-Set Closure Specialist	Closure/missingness for borders/stock, parent/subsidiary and adjacency checks.	Mixed
M16	Atomic Consensus Engine	Independent support, same-view repeat control, numeric/set consensus.	No
M17	Specialist Verifier Suite	Relation-specific classification prompts over same A/B/C or richer temporary labels.	Yes
M18	Bidirectional/Counterfactual Checks	Reverse relation, key-condition reconstruction, contrastive negatives.	Yes
M19	Coverage Gap/Missingness	Facet gaps, novelty rate, singleton ratio, saturation, disagreement.	No
M20	Relation Budget Scheduler	Reserve discovery/verification/freshness budget by relation.	No
M21	Expected-Value Micro-Planner	Pick next action by expected verified gain, uncertainty reduction, redundancy and cost.	No

5 Layer 1 – M9 Risk/Difficulty Profiler

M1 knows the relation type; M9 estimates *instance difficulty*. The router is not a learned classifier. It reads the relation, subject surface form, initial graph, and early-return signals to produce the vector:

$$Q = (q_{card}, q_{temp}, q_{num}, q_{open}, q_{amb}, q_{novel}, q_{verify}).$$

For example, awards tend to have high q_{open} ; capacity has high q_{num} ; death/stock have high q_{temp} ; borders often have low q_{amb} after direct+compass views. M9 selects a branch and compute envelope only; it never contains a factual answer.

5.1 Relation routing

Table 3: Default routing and secondary paths.

Relation	Primary specialist	Secondary modules
countryLandBordersCountry personHasCityOfDeath	M15 Small-Set Closure M14 Null/Temporal	M18 reverse check for singleton/territory risk M11 pseudo-memory; M18 key-condition; cross-model freshness
hasCapacity	M12 Numeric	M17 numeric verifier; M18 contrast attendance/capacity
awardWonBy	M13 Large-Set	M11 query specification; M19 missingness; M20 reserved verify
companyTradesAtStockExchange	M15 Small-Set + temporal subroutine	M14 freshness; M18 parent/subsidiary contrast
hasArea	M12 Numeric	M17 total-vs-land area verifier; cross-unit cycle

6 Layer 1 – M10 Prompt Program Compiler

6.1 From prompt engineering to executable prompt programs

APE treats instructions as programs to be searched; OPRO uses natural-language optimization; MIPRO optimizes prompts for multi-stage LM programs; AMPO grows multi-branched prompts from failure patterns [13, 14, 15, 16]. COVER adopts a rule-safe variant:

1. define a fixed semantic contract;
2. generate a candidate prompt set on TRAIN;
3. evaluate end-to-end with the official metric plus diagnostics;
4. retain versioned prompts in config without changing weights;
5. freeze one prompt package before each validation submission.

We do not use learned surrogates from MIPRO/PROMST; we only adopt module-aware optimization and discrete search.

6.2 PromptProgram structure

```
PromptProgram(
  relation_id,
  positive_contract,
  negative_contract,
  keyword_bundles,
  step_back_template,
  direct_templates,
  facet_templates,
  pseudo_memory_templates,
  verifier_templates,
  output_grammar,
  independence_groups,
  stop_strings,
)
```

6.3 Relation keyword specification

Here, “keyword” means lexical steering, **not** a query sent to a web/search engine.

Table 4: Example keyword bundles and negative anchors.

Relation	Positive keyword bundles	Negative/contrast anchors
Borders	land border, land boundary, adjacent, frontier, neighboring state, enclave boundary	maritime-only, nearby, sea boundary, disputed-but-no-contact
Death city	died, death place, city/locality of death, passed away, deceased	birthplace, residence, hospital name without city, country/region, living
Capacity	maximum spectator capacity, official capacity, seating+standing, current/historical configuration	record attendance, average attendance, field dimensions, seats-only if total asked
Award	winner, laureate, recipient, honoree, year, category, medal/prize recipient	nominee, winning work instead of creator, similarly named award, rescinded award
Stock exchange	listed, traded, exchange, ticker/listing, primary/secondary listing	parent only, subsidiary only, private company, index not company, delisted without current listing
Area	total area, km2, square miles, hectares, water+land total	land area only, population, coastline, length

6.4 Step-back prompt layer

Step-Back Prompting improves reasoning by first extracting higher-level concepts/principles [12]. In COVER, the step-back phase is restricted to producing a *search specification*, not a factual answer:

Given the relation contract, list the semantic subfacets, likely ambiguity classes, units/time slices, and exclusion rules needed to retrieve this fact from parametric memory. Do not answer the factual query.

The output becomes the branch plan for M11–M15.

7 Layer 1 – M11 Closed-book Parametric Retrieval

7.1 Why call it “parametric retrieval”?

HyDE and Query2Doc show that LLM-generated pseudo-documents/query expansions can enrich a query representation; Rewrite-Retrieve-Read emphasizes query rewriting; FLARE emphasizes low-confidence retrieval triggers; Self-RAG and LLaRetrieval emphasize retrieve/critique/verify loops [22, 23, 24, 25, 21, 26]. Since external documents are forbidden, COVER replaces corpus retrieval with *independent parametric-memory probes*.

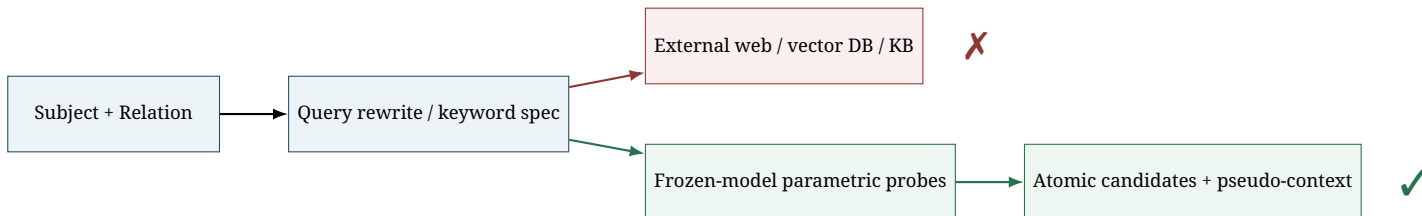


Figure 2: RAG literature informs the query/retrieval policy, but external retrieval is hard-disabled; COVER only calls frozen models and stores per-query ephemeral evidence.

7.2 Three probe families

Pseudo-memory view. The model writes a relation-focused memory sketch; the parser only extracts candidates. The sketch does not itself create a support edge.

Self-ask decomposition. Self-ask decomposes a query into follow-up subquestions [27]; in COVER, the follow-ups are answered by a frozen model, not by a search engine.

Query rewrite loop. If action yield is low or disagreement is high, M11 rewrites the query using a missing facet/negative constraint instead of merely changing the random seed.

Evidence hygiene

Pseudo-context, generated explanations, and chain-of-thought are never passed verbatim into the verifier. They are acquisition artifacts. The Evidence Graph records only candidate occurrence, independence group, raw logit diagnostics, and verifier results. This prevents self-confirmation and preserves the blind-verification invariant.

8 Layer 2 – M12 Numeric Specialist: hasCapacity and hasArea

Validation shows that hasCapacity has high precision but extremely low recall. This is a signature of over-abstention more than pure hallucination. M12 reframes a numeric query as a consensus problem.

8.1 Numeric multi-probe

The independence groups are:

- exact-quantity direct;
- contrastive definition (capacity vs attendance; total vs land area);
- cross-unit/cross-format;
- historical/current configuration where contract permits;
- candidate-free re-elicitation.

8.2 Canonicalization and clustering

Each observation x_i is deterministically converted to a canonical unit. Two values belong to the same cluster if

$$\delta(x_i, x_j) = \frac{|x_i - x_j|}{\max(|x_i|, |x_j|, \epsilon)} \leq \tau_{cluster, r}.$$

The dominant cluster C^* has representative

$$\hat{x} = \text{median}(C^*), \quad D_{num} = \frac{\text{MAD}(C^*)}{|\text{median}(C^*)| + \epsilon}.$$

The state stores independent_support, total_support, dispersion, verifier VALID/INVALID/UNKNOWN evidence, and hard-definition violations.

8.3 UNKNOWN is not a contradiction

Contextual calibration reduces label bias [19]. However, verifier label C=UNKNOWN only indicates insufficient parametric confidence. The new numeric decision rule is:

$$\text{ACCEPT}(x) \Leftarrow I(C^*) \geq k_r \wedge D_{num} < \tau_{disp, r} \wedge \neg \text{HARDINVALID}(x)$$

even when the verifier returns UNKNOWN, provided there is no strong INVALID signal and independent consensus is sufficiently strong. This is the most important semantic change for capacity.

8.4 Contrastive numeric verification

Instead of asking “Is 25,000 correct?”, M17 can present a small choice set: the candidate cluster center, the nearest competing cluster, and UNKNOWN. Pairwise/contrastive burden is lower than listwise ranking, consistent with findings that pairwise prompting can simplify ranking for LLMs [35].

9 Layer 2 – M13 Large-Open-Set Specialist: awardWonBy

Award recovery is a set-reconstruction problem, not one-shot list generation. ASC shows that merging atomic subparts across multiple samples can outperform selecting a single generation [5]; ReWiSe and LM-KBC divide-and-conquer show that relation-wise repeated probing/decomposition is useful for KB completion [2, 3].

9.1 Facet decomposition

M13 runs a direct seed query and then creates generic, non-factual facets:

- temporal slices / eras;
- recipient type (person, organization, creator where contract allows);
- official category/discipline dimension when the award defines categories;
- geography only when semantically appropriate;
- missingness facet: which part of the candidate graph is still underrepresented.

9.2 Atomic support score

For candidate o :

$$S_{award}(o) = w_I I(o) + w_V \tilde{p}_V(o) + w_X X(o) - w_C C(o) - w_R R_{near}(o),$$

where $I(o)$ is the number of independence groups, $X(o)$ is cross-model/candidate-free support, and R_{near} is near-miss risk. Same-view stochastic repeats increase only total support, not I .

9.3 Compute reservation

Large-set discovery must not consume the entire budget. For budget B_r :

$$B_r = B_{seed} + B_{facet} + B_{verify} + B_{reverse} + B_{reserve},$$

where B_{verify} is a hard reservation that discovery cannot spend. This directly addresses the observed failure in which award queries generate tens or hundreds of candidates while almost none are verified.

9.4 Tiered set pruning

- Tier A: multi-view independent support → cheap acceptance/spot-check.
- Tier B: single-view but plausible → blind verifier.
- Tier C: high-risk near-miss → adversarial/contrastive verifier.
- Tier D: weak singleton + no corroboration → prune unless missingness remains high.

10 Layer 2 – M14 Null/Temporal Specialist: personHasCityOfDeath and freshness

A zero-or-one relation must separate two questions: “does an object exist?” and “which object is it?” Self-correction literature warns that merely asking a model to revise its previous answer is often insufficient [30]; key-condition verification suggests targeted verification is more effective than generic self-reflection [31].

10.1 Two-stage gate

Stage A: existence/death-status. Independent prompts predict {living, deceased, unknown}. No city is inferred until the gate has sufficient evidence.

Stage B: locality acquisition. If deceased/non-empty is plausible, run direct locality, biography-locality, birth-vs-residence contrast, and candidate-free recall.

10.2 Temporal freshness branch

The gold data is updated through 1 July 2026 [1]. For death, stock, and recent awards, model freshness is a distinct risk dimension. M14 may trigger **cross-family fresh recall**: a smaller/fresher model can generate candidates on one branch, after which the other model blind-verifies them. No factual web lookup is used.

10.3 NULL evidence

An empty output is accepted only when there is independent null evidence; “no candidate was generated” is not automatically equivalent to “gold is empty”. The state separates:

$$E_{null} = \{\text{living support, no-known-locality support, failed-recall only}\}.$$

Failed recall receives very little weight.

11 Layer 2 – M15 Small-Set Closure Specialist

Borders and stock exchanges usually have small cardinality, so the objective is high-precision closure.

11.1 Borders

Borders already reach 0.9531; the default policy must therefore be **minimal-change**. Use direct + geographic decomposition + reverse checks for singleton/territory ambiguity. Do not increase compute if the set is already stable.

11.2 Stock exchange

Stock requires a public-listing gate, company-itself checks, primary/secondary/dual listing handling, and temporal status. An abnormally long candidate list triggers parent/subsidiary/index confusion filters. M14s freshness branch may be invoked as a subroutine.

11.3 Closure test

Given accepted set A_t , the missingness probe returns new candidates N_t . Stop when

$$|N_t| = 0, \quad J(A_t, A_{t-1}) > \tau_J,$$

and no high-risk singleton remains.

12 Layer 3 – M16 Atomic Consensus Engine

Self-Consistency selects an answer mode [4]; ASC merges atomic facts [5]. COVER needs a relation-aware variant: an atomic candidate graph rather than list-level voting.

12.1 Independent consensus

Each evidence event e has an `independence_group`. For group g :

$$q_g(o) = \max_{e \in g} \text{support}(e, o),$$

without summing all repeats. The candidate support vector is:

$$\phi(o) = (F, L, X, C, U, I, D, \text{cost}, \text{risk}).$$

12.2 Semantic disagreement

No additional embedding model is required by default. Entity/string relations use normalized candidate sets; numeric relations use clusters. An optional semantic-equivalence judge is enabled only if the model budget permits it and alias over-merging remains controlled.

13 Layer 4 – M17 Specialist Verifier Suite

A single generic A/B/C prompt is too coarse for all relations. M17 preserves CoVe’s blind-verification invariant [6], but each relation gets its own verifier contract.

Table 5: Specialist verification contracts.

Specialist	Verifier question	Typical hard negative
Numeric	Does the candidate satisfy the exact quantity definition?	attendance vs capacity; land vs total area
Award membership	Is the candidate a recipient of this exact award?	nominee, work, similar award, rescinded
Null/temporal	Are existence/death status/locality conditions consistent?	living, birthplace, residence, country
Stock	Company itself currently/contractually listed on exchange?	parent/subsidiary/index/private/delisted
Border	Physical land contact?	maritime-only/nearby territory

LLMs may hallucinate during generation yet still perform well as verifiers; NAACL 2024 shows that generator quality and verifier quality need not coincide [28]. Therefore model-role bake-offs must measure enumeration and verification separately.

13.1 Verifier bias controls

In addition to contextual calibration [19], test template-order/label bias and label-order swaps in calibration controls. ACL 2024 documents positional bias in LLM-as-evaluator settings [29]; COVER avoids free-form judge scores and uses fixed labels, but still logs template disagreement.

14 Layer 4 – M18 Bidirectional and Counterfactual Verification

M18 creates genuinely new evidence instead of issuing a generic “think again” instruction.

- **Reverse check:** candidate $o \rightarrow$ subject s when the relation supports a meaningful reverse question.
- **Key-condition reconstruction:** mask the subject/candidate condition and ask the model to recover it; use the resulting consistency signal, following targeted condition verification ideas [31].
- **Counterfactual pair:** compare a true-looking candidate against a near-miss class generated from the contract, not from external facts.
- **Candidate-free recall:** do not show the candidate; if it appears naturally in an independent probe, increase X .

14.1 Optional DoLa/factual decoding adapter

DoLa (ICLR 2024) contrasts later-layer and earlier-layer logits to improve factuality without training [20]. COVER treats it as an optional experimental adapter, not a core component: enable it only on compatible text-only generation paths after parameter/runtime audit, and retain it only if validation ablation shows a gain. Do not apply DoLa to the A/B/C verifier if it changes calibration semantics.

15 Layer 5 – M19 Coverage Gap and Missingness Estimator

The previous proposal used RCSE rather than treating Chao2 as a cardinality oracle. The upgraded version constructs a non-neural ensemble:

$$R_t = w_1 \text{noveltyRate}_t + w_2 \text{singletonRatio}_t + w_3 \text{facetGap}_t + w_4 \text{disagreement}_t + w_5 \text{unresolvedMass}_t.$$

Set-valued relations use incidence statistics as heuristics; numeric relations use cluster stability; null-single relations use competing-state uncertainty.

15.1 Facet coverage map

Each relation has a facet registry. M19 marks facets as {covered, weak, unexplored, exhausted}. For awards, missingness asks not only “who else?” but which facet lacks support. For stock, primary/secondary/dual listings are semantic facets. For borders, geographic decomposition defines the facets.

16 Layer 6 – M20 Relation Budget Scheduler

Adaptive-Consistency shows that fixed sample counts waste compute once agreement is clear [7]; ICLR 2025 inference-scaling work shows that model size and inference strategy should be optimized jointly under a compute budget [32, 33]. M20 allocates budget by relation and reserves budget by action class.

Table 6: Qualitative budget policy; concrete values are calibrated on TRAIN.

Relation	Discovery	Verification	Special reserve
Borders	low	low/spot	reverse singleton only
Capacity/Area	medium multi-probe	medium	cross-unit/contrast
Awards	high but capped	hard-reserved high	missingness + reverse
Death	medium	medium-high	freshness/candidate-free
Stock	medium	medium	freshness + parent/sub-sidiary

Budget accounting must be cache-aware and precharge before every neural call. No action may exceed the hard cap.

17 Layer 6 – M21 Expected-Value Micro-Planner

SELF-DISCOVER chooses reasoning structures by task [8]; LATS searches over reasoning/action/planning [11]; Tree of Thoughts shows that deliberate branching can outperform left-to-right decoding on search-heavy tasks [10]. COVER does not require full MCTS. Instead it uses **1–2 step micro-lookahead** over a small action space.

Action utility:

$$U_t(a) = \alpha \widehat{G}_{verified}(a) + \beta \widehat{\Delta R}(a) + \gamma \widehat{\Delta H}(a) - \delta \widehat{Cost}(a) - \eta \widehat{Redundancy}(a) - \kappa \widehat{FP}(a).$$

The estimates come from relation-specific historical bins on TRAIN; no neural policy is trained. M21 selects $a^* = \arg \max U_t(a)$ if $U_t(a^*) > \tau_{continue}$; otherwise it returns STOP.

17.1 Policy examples

- Capacity: three independent values form a tight cluster + Qwen UNKNOWN $\rightarrow$ accept rather than wasting verifier loops.
- Award: novelty remains high but verification reserve is unused $\rightarrow$ verify the shortlist before opening another facet.
- Border: direct/compass sets are stable $\rightarrow$ stop early.
- Death: no candidate, but null evidence is only failed recall $\rightarrow$ run the fresh/candidate-free branch before returning empty.

18 Model plane – pretrained model portfolio and role routing

18.1 Principles

No model is enabled because of benchmark marketing. Every profile must have an exact model ID/revision, published total parameter count, license, tokenizer/runtime compatibility, real-weight smoke test, and validation bake-off. The challenge counts MoE models by total published parameters; quantization does not reduce that count [1].

Table 7: Model profiles worth baking off. “Provisional” means the exact total neural components must be audited before use.

Profile	Published size	Budget status	Role / remarks
Current: Mistral Small 3.2 + Qwen3.5-4B	24.011B + 4.660B	PASS 28.67B	Cross-family; runtime already validated; retain as anchor baseline. Mistral 3.2 emphasizes instruction-following/repetition fixes [36]; the smaller Qwen acts as verifier/cross-recall model.
Qwen3-30B-A3B single	30.5B total / 3.3B active	PASS by published total if no extra neural model	MoE, but the challenge counts all 30.5B; supports many test-time samples with low active compute, at the cost of losing cross-family verification [37].
Qwen3-14B + Phi-4 14B	14.8B + 14B	approx. 28.8B; audit revisions	Dense cross-family pairing; Phi-4’s public-data cutoff of June 2024 makes temporal relations riskier [38, 39].
Qwen3.6-27B single	27B language component	PROVISIONAL	2026 family, attractive for knowledge freshness/stability; the checkpoint includes a vision encoder, so full neural total/loaded components must be audited before enabling [40].
Qwen3.5-27B single	27B language component	PROVISIONAL	Long context, 2026 family; multimodal total requires conservative auditing [41].
Gemma 3 27B single	27B-class multimodal	PROVISIONAL	Alternative family; gated Gemma terms and multimodal parameter/runtime accounting require audit [42].
Mistral Small 3.2 + Phi-4-mini	24.011B + 3.8B	approx. 27.8B	Cross-family verifier alternative, but Phi-mini has a public-data cutoff of June 2024 [43].

18.2 Model roles need not be fixed

In the current profile, Mistral is the default enumerator and Qwen is the verifier. The upgrade allows static relation-wise role swaps:

- static facts: prefer the model with the highest recall on TRAIN;
- temporal relations: allow a fresher family to run one candidate-free/cross-recall branch;
- verification: choose the model with the best calibrated A/B/C discrimination, not necessarily the strongest generator [28].

The router is a configuration table, not a learned model.

19 Prompt optimization pipeline

19.1 Method sources

APE searches instruction candidates [13]; OPRO uses previous solutions/scores to propose improved prompts [14]; MIPRO targets multi-stage LM programs [15]; AMPO learns multi-branch prompt struc-

ture from failure patterns [16]; BPO and adversarial ICL show that input-side optimization can improve models without weight updates [17, 18]. COVER adopts only the training-free/discrete-search-compatible components.

19.2 COVER Prompt Lab

Each specialist has an offline Prompt Lab:

1. collect TRAIN error clusters by relation;
2. generate 5–20 prompt variants from a fixed meta-prompt;
3. run small deterministic TRAIN slice + full TRAIN candidate finalists;
4. score official F1 + candidate count + empty rate + calls/token;
5. prune prompt branches that cause drift;
6. freeze prompt package version;
7. run full validation and submit to the leaderboard once per architectural revision.

Validation is not used as an inner loop for dozens of micro-threshold sweeps; however, under the current development policy, every meaningful module revision still receives a leaderboard submission to measure external gain.

19.3 Prompt invariants

- contract-first, output grammar-last;
- one factual objective per prompt;
- no chain-of-thought transfer from generator to verifier;
- explicit hard negatives;
- independent branches must differ in semantic decomposition, not merely wording/seed;
- prompt version + hash are logged for every call.

20 End-to-end flow by relation

20.1 countryLandBordersCountry

1. M0/M1 compile SMALL_SET physical-land-border contract.
2. M9 estimates risk and typically routes to the M15 fast path.
3. Direct + compass/region views; normalize countries/territories.
4. Only singleton/territory ambiguity invokes M18 reverse checking.
5. M19 measures set stability/closure; M21 stops early.

The objective is to avoid damaging the existing F1 of 0.9531.

20.2 personHasCityOfDeath

1. M14 existence/death-status gate.
2. If non-empty is plausible: locality acquisition + pseudo-memory/candidate-free branch.
3. If temporal risk is high: Qwen fresh-recall branch + cross-family verification.
4. M17 distinguishes death city from birthplace/residence/country.
5. Empty is returned only when independent null evidence is sufficiently strong.

20.3 hasCapacity

1. M12 multi-probe maximum spectator capacity/current-historical/contrast attendance.
2. Canonicalize integers; cluster values.
3. A strong low-dispersion independent cluster may survive verifier UNKNOWN.
4. M17 performs exact-quantity verification; only INVALID is strong contradiction evidence.
5. Final robust representative, evaluator tolerance 5%.

20.4 awardWonBy

1. Seed direct set.
2. M10/M11 compile facet queries; M13 divide-and-conquer.
3. M16 atomic union with independence support.
4. M20 reserve verification budget; shortlist before more exploration.
5. M17/M18 eliminate nominee/work/similar-award errors.
6. M19 facet missingness + novelty saturation; M21 chooses continue/stop.

20.5 companyTradesAtStockExchange

1. M15 public-listing gate + listing-type facets.
2. M14 temporal/freshness subroutine if listing status uncertain.
3. M18 parent/subsidiary counterfactual and company-itself verification.
4. Small-set closure; avoid large cross-model candidate explosions.

20.6 hasArea

1. M12 asks total area in km² plus independent cross-unit/definition views.
2. Deterministic conversion and numeric clustering.
3. M17 checks total vs land-only definition.
4. Strong consensus can accept despite UNKNOWN; competing clusters trigger one additional branch.

21 Software architecture

21.1 Extended repository layout

```
src/cover_kbc/
  compiler/          # M0 semantics + relation programs
  routing/           # M1 + M9 risk/profile routing
  prompts/           # M10 PromptProgram + versioned templates
  parametric/        # M11 pseudo-memory/self-ask/rewrite probes
  specialists/
    numeric.py       # M12
    large_set.py     # M13
    null_temporal.py # M14
    small_set.py     # M15
  evidence/          # M3 + M16 graph/consensus
  verification/      # M4 + M17 + M18
  coverage/          # M6 + M19
  control/           # M7 + M20 + M21
  models/            # frozen runtimes, role swap, tokenizer adapters
  finalization/      # M8 relation-aware writer
configs/
  relations/
  prompts/
  policies/
  model_profiles/
experiments/
reports/
tests/
```

21.2 Interface invariants

- specialist modules emit evidence events and never bypass the graph;
- all neural calls go through runtime accounting;
- all relation/model/prompt decisions versioned in config;
- no network-facing factual source reachable from inference path;
- benchmark immutable;
- output always official JSONL contract.

22 Upgraded inference algorithm

Algorithm 1 COVER-KBC Super-System for one query

Require: subject s , relation r , relation budget B_r

```

1:  $K \leftarrow \text{COMPILERELATION}(r)$ ;  $P \leftarrow \text{COMPILEPROMPTPROGRAM}(K)$ 
2:  $Q \leftarrow \text{PROFILERISK}(s, K)$ ;  $E \leftarrow \text{ROUTE SPECIALIST}(K, Q)$ 
3:  $G \leftarrow \text{INITGRAPH}(s, r, K)$ ;  $B \leftarrow \text{RESERVEBUDGET}(B_r, K, Q)$ 
4: for all  $v \in \text{MANDATORYVIEWS}(P, E)$  do
5:    $y, z, c \leftarrow \text{RUNFROZENMODEL}(v, s, r)$ 
6:    $O \leftarrow \text{PARSENORMALIZE}(y, K)$ ;  $G \leftarrow \text{ADD DISCOVERY}(G, O, v, z, c)$ 
7: end for
8: while  $\text{BUDGETLEFT}(B)$  do
9:    $G \leftarrow \text{ATOMICCONSENSUS}(G, K)$ 
10:   $U \leftarrow \text{UNRESOLVED}(G, K)$ ;  $R \leftarrow \text{COVERAGEGAP}(G, K)$ 
11:  if  $\text{SHOULDSTOP}(G, R, B, K)$  then
12:    break
13:  end if
14:   $A \leftarrow \text{LEGALACTIONS}(G, R, B, K, E)$ 
15:   $a \leftarrow \arg \max_{a \in A} \text{EXPECTEDUTILITY}(a, G, R, B)$ 
16:  if  $a = \text{STOP}$  then
17:    break
18:  end if
19:   $G, B \leftarrow \text{EXECUTEANDRECORD}(a, G, B, K, P, E)$ 
20: end while
21: return  $\text{FINALIZE}(G, K)$ 
  
```

23 Experimental plan and validation leaderboard loop

23.1 Locked baseline

Current validation baseline: overall 0.4230. Every subsequent submission must store commit SHA, config hash, prompt-package version, model revisions, and artifacts.

23.2 Incremental module ladder

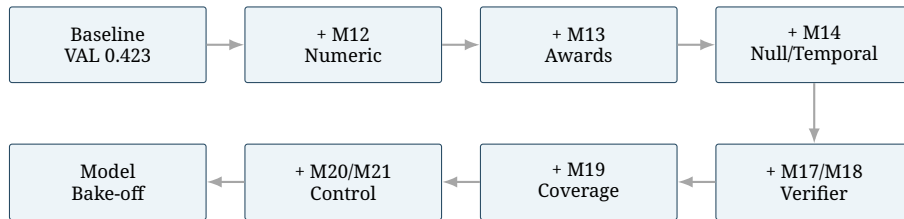


Figure 3: Each architectural increment receives TRAIN analysis + a full VAL run + leaderboard submission; keep only modules with clear gain or diagnostic value.

Table 8: Incremental module/layer implementation roadmap.

Step	Change	Target relation	Gate for keeping the module
S0	Baseline frozen M0–M8	all	VAL 0.4230 anchor
S1	M12 Numeric Specialist + M16 numeric consensus	capacity, area	capacity recall increases clearly, precision does not collapse; overall non-negative
S2	M13 Large-Set Specialist + verify reservation	awards	award F1 increases, candidate explosion decreases, verification coverage increases
S3	M14 Null/Temporal	death + stock freshness	death recall/empty calibration improves; stock does not regress

Step	Change	Target relation	Gate for keeping the module
S4	M17 Specialist Verifier	numeric/award/death/stock	lower UNKNOWN misuse, better candidate precision/recall frontier
S5	M18 Bidirectional checks	high-risk cases	gain on near-miss subsets at bounded calls
S6	M10/M11 Prompt Program optimization	all targeted	per-relation gain under frozen architecture
S7	M19 Coverage Gap	award/stock/border	next-call novelty prediction useful; better stop/continue decisions
S8	M20 Relation Budget Scheduler	all	higher F1 or same F1 with fewer wasted calls
S9	M21 Expected-Value Planner	all	best F1–cost frontier; no instability
S10	Optional DoLa adapter	factual branches	keep only if controlled validation gain
S11	Model profile bake-off	all	exact budget PASS + highest validation score + reproducible runtime

23.3 Metrics

Beyond macro P/R/F1 and leaderboard score:

- per-relation F1; empty precision/recall; prediction-count distribution;
- calls/query, prompt/generated tokens, latency;
- verifier label distribution, margin/entropy, disagreement;
- candidate provenance depth and independent-support histograms;
- numeric cluster accuracy/dispersion;
- award shortlist precision and verified coverage;
- next-action verified yield and the F1–cost frontier.

23.4 Validation discipline

The workflow intentionally uses the leaderboard frequently. To keep the paper scientifically defensible:

1. every leaderboard submission must correspond to an architectural revision, not micro-threshold spam;
2. threshold/prompt candidate search should primarily use TRAIN/internal folds;
3. log all validation submissions, including regressions;
4. report the chronological ablation path in the final paper rather than cherry-picking only best runs;
5. freeze code/config/model revisions before TEST.

24 New ablation ladder

Table 9: Ablation matrix for the system paper.

System	PromptProg	Spec.	Consensus	Verify+	Coverage	Budget	Planner	ModelRoute
Baseline M0–M8	–	–	base	base	RCSE	base	–	fixed
+ Numeric	–	Num	✓	base	RCSE	base	–	fixed
+ Award	–	Num+Award	✓	base	RCSE	reserve	–	fixed
+ Null/Temp	–	+Null	✓	base	RCSE	reserve	–	fixed
+ Specialist Verify	–	✓	✓	✓	RCSE	reserve	–	fixed
+ Prompt Programs	✓	✓	✓	✓	RCSE	reserve	–	fixed
+ Coverage Gap	✓	✓	✓	✓	✓	reserve	–	fixed
+ Active Control	✓	✓	✓	✓	✓	✓	✓	fixed
Full + model routing	✓	✓	✓	✓	✓	✓	✓	✓

25 Relation to prior work and novelty boundary

25.1 What is inherited

- Self-Consistency/ASC: sample diversity and atomic merging [4, 5].
- ReWiSe/LM-KBC DaC: relation-wise aggregation and decomposition [2, 3].
- CoVe/ProCo: independent/targeted verification [6, 31].
- Contextual calibration: label-bias correction [19].
- SELF-DISCOVER, ReAct, ToT, LATS: structured reasoning/action decomposition and search [8, 9, 10, 11].
- Adaptive Consistency + inference scaling: non-uniform test-time compute [7, 32, 33].
- APE/OPRO/MIPRO/AMPO/BPO: prompt optimization/programming [13, 14, 15, 16, 17].
- RAG family: query rewriting, active retrieval, retrieval verification; only the control principles are used, with no external retrieval [21, 22, 24, 25, 26].
- Entity Set Expansion via LM probing: positive/negative semantic classes to reduce drift [34].
- DoLa: optional layer-contrastive factual decoding [20].

25.2 COVER-KBC claims

We do not claim to have invented every individual block. The system-level claims should be:

1. **Relation-Typed Active Evidence Acquisition:** fixed relations are compiled into executable specialist programs with relation-specific discovery, verification, coverage, and stopping semantics.
2. **Independence-Aware Calibrated Atomic Evidence:** candidate-level evidence distinguishes structural independence, model family, blind label logits, contradiction, and prompt disagreement.
3. **Coverage/Uncertainty-Guided Compute Control:** action budgets are allocated by expected marginal verified yield, residual coverage, and risk rather than by fixed sample count.
4. **Challenge-legal RAG abstraction:** query-rewriting/active-retrieval principles are translated into closed-book parametric retrieval with no external factual source.

One-sentence description of the upgraded system

COVER-KBC compiles each relation into a routed specialist inference program, actively retrieves candidate evidence from frozen parametric memory through structurally independent prompt branches, verifies atomic candidates with calibrated and counterfactual checks, estimates unresolved coverage, and spends further test-time compute only on the action with the highest expected verified gain.

26 Technical risks and fallbacks

Table 10: Risks of the super-system.

Risk	Consequence	Fallback / invariant
Prompt branches correlated	fake consensus	independence groups + branch audits; same-view repeats do not increase independent support
Pseudo-memory confirms	hallucination amplification	pseudo-context is not evidence; blind verifier never sees generator reasoning
RAG rule violation	disqualification	network/DB retrieval hard-disabled; compliance tests
Numeric UNKNOWN veto	low recall	UNKNOWN=uncertainty; strong cluster can survive
Award candidate explosion	precision/compute collapse	shortlist + hard verification reservation

Risk	Consequence	Fallback / invariant
Death empty overprediction	recall collapse	distinguish failed recall from positive NULL evidence
Verifier bias	wrong accept/reject	contextual calibration + template/order disagreement
Too many modules	diffuse paper narrative	paper claims 3–4 abstractions; specialists are instantiations/ablations
Planner instability	nondeterministic regressions	rule-based utility, bounded lookahead, exact config/versioning
Model budget ambiguity	rule risk	fail-closed parameter audit before enable
Fresh model tokenizer/runtime bugs	broken inference	real-weight R0 smoke per profile, checkpoint-native tokenizer adapters
Validation overfitting	misleading leaderboard gains	chronological submissions + TRAIN inner-loop + frozen revision per VAL

27 Execution roadmap

1. **M12 Numeric:** code first, because capacity 0.09 is the clearest correctable failure. Run TRAIN + full VAL + submit.
2. **M13 Award:** add divide-and-conquer, atomic shortlist, reserved verification. Run VAL + submit.
3. **M14 Null/Temporal:** fix death empty behavior and add freshness cross-recall. Run VAL + submit.
4. **M17/M18 Verification:** specialist prompts + targeted counterfactual/reverse checks. Run VAL + submit.
5. **M10/M11 Prompt Programs:** automated discrete prompt search on TRAIN, versioned branch compiler. Run VAL + submit.
6. **M19 Coverage:** facet-gap ensemble and missingness. Run VAL + submit.
7. **M20/M21 Control:** budget reservation + expected-value micro-planner. Run VAL + submit.
8. **Optional factual decoding:** DoLa behind flag; keep only with measured gain.
9. **Model bake-off:** current profile vs legal alternatives; exact parameter/runtime audit before each.
10. **Final freeze:** code, config, prompts, model revisions, evaluator hash, then official TEST when submission phase is available.

28 Conclusion

The previous proposal was correct to frame KB completion as active set discovery rather than answer generation. This upgrade turns that abstraction into a routed, layered, specialist-driven system. RAG papers are not copied into external retrieval because the rules prohibit it; instead, query rewriting, active retrieval, and verification are reinterpreted as closed-book parametric-memory elicitation. Prompt optimization becomes a first-class engineering layer. Numeric, large-set, null/temporal, and small-set relations pass through different specialists. Evidence still converges into one shared graph, the verifier remains blind, the controller remains deterministic/auditable, and every neural component remains frozen.

The objective is not to create a complicated “agent maze” for appearance’s sake. Every new module must address an observed failure mode and survive TRAIN + full validation + leaderboard ablation. If a module does not increase F1 or improve the F1-cost/diagnostic frontier, it is removed. In this way, COVER-KBC Super-System can grow in capability while remaining tight, reproducible, and scientifically coherent.

A Illustrative prompt templates

A.1 A.1 Step-back query specification

System:

You are compiling a retrieval plan for a closed-book factual relation.
Do NOT answer the factual query.
Do NOT use external sources.

Relation contract: <CONTRACT>

Subject: <SUBJECT>

Return:

1. positive semantic anchors
2. hard-negative anchors
3. useful independent subfacets
4. unit/time distinctions if relevant
5. a strict output grammar for the next factual probe

A.2 A.2 Pseudo-memory acquisition

Use only your parametric knowledge.

Write a compact memory sketch focused only on <RELATION> for <SUBJECT>.

Separate uncertain details from confident details.

Do not infer from unrelated facts.

The sketch is only used to propose candidates; it is not evidence.

A.3 A.3 Numeric capacity contrast

Subject venue: <VENUE>

Quantity: highest published maximum spectator capacity.

Exclude average attendance and record attendance.

Distinguish seated-only capacity from total spectator capacity.

Return exactly one integer, or UNKNOWN if no factual value is available.

A.4 A.4 Award facet

Award: <AWARD>

Already seen candidates: <SHORT_LIST>

Facet specification: <PERIOD_OR_RECIPIENT_TYPE>

Return only recipients of this exact award from the specified facet.

Do not return nominees, works when the recipient should be the creator,
or recipients of similarly named awards. Do not repeat existing candidates.

A.5 A.5 Blind verifier

You verify one candidate against an exact relation contract.

You do not see the generator's reasoning.

Use only your parametric knowledge.

Subject: <SUBJECT>

Relation definition: <CONTRACT>

Candidate: <CANDIDATE>

Hard negatives: <NEGATIVES>

A = VALID

B = INVALID

C = UNKNOWN / insufficient parametric knowledge

Return one label only.

A.6 A.6 Key-condition reconstruction

Given the relation contract and candidate <CANDIDATE>, determine whether
<SUBJECT> is consistent with being the corresponding subject.

Do not rely on the previous generation. Return A/B/C only.

B Relation keyword programs

Table 11: Lexical programs for M10; these are prompt-steering devices, not external search.

Relation	Query facets	Rewrite triggers
Borders	direct; compass/region; reverse; territory status	duplicate/sea-border words; singleton; unexplored direction
Death	status; locality; biography locality; contrast birth/residence; freshness	all-empty; living/deceased disagreement; country-level candidate
Capacity	max/current; renovation; seated-vs-total; attendance contrast	competing numeric clusters; UNKNOWN; non-integer surface
Award	seed; temporal; recipient type; category; missingness	candidate explosion; one-off tail; uncovered facet; high near-miss rate
Stock	listing gate; primary; secondary/dual; temporal status; company-itself	many exchanges; parent/subsidiary terms; delisting uncertainty
Area	total-area direct; cross-unit; land-vs-total contrast	unit conflict; cluster split; land-area cue

C Module I/O contracts

Table 12: I/O contracts for module-by-module implementation.

Module	Input	Output	Neural
M9 Risk Profiler	QuerySpec + early graph	RiskProfile + route hints	No
M10 Prompt Compiler	contract + profile + prompt version	PromptProgram	No
M11 Parametric Retrieval	PromptProgram + state	GenerationRecords/pseudo-memory candidates	Yes
M12 Numeric Specialist	numeric QuerySpec + evidence	NumericClusterState + actions	Mixed
M13 Large-Set Specialist	set QuerySpec + graph	facet plan + shortlist	Mixed
M14 Null/Temporal	query + temporal/null state	existence/locality/freshness events	Mixed
M15 Small-Set Closure	query + set state	closure/reverse actions	Mixed
M16 Consensus	evidence events	candidate consensus states	No
M17 Specialist Verifier	candidate + specialist contract	label distributions + verdict	Yes
M18 Bidirectional Check	candidate + relation	independent support/contradiction	Yes
M19 Coverage Gap	graph + facet registry	residual/gap state	No
M20 Budget Scheduler	relation + risk + remaining budget	reserved envelopes	No
M21 Micro-Planner	full state + legal actions	next action / STOP	No

D Model bake-off checklist

Before adding a new profile:

1. verify license and exact immutable revision;
2. derive conservative published total parameter count of every inference-time neural component;
3. fail if total > 32B or ambiguous;
4. verify tokenizer/chat template and label tokenization;

5. real-weight one-call generation + one-call scoring smoke;
6. run identical TRAIN slice, then full VAL only after runtime pass;
7. compare per-relation enumeration recall, verifier discrimination, empty behavior, cost and overall F1;
8. preserve previous profile as reproducible baseline.

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